

Workshop Practice Lab:

Workshop Practice Lab is a course that introduces students to the basic skills and techniques used in various workshop processes. The course covers a range of topics, including:

1. **Safety:** Students learn about the importance of safety in the workshop, including the use of personal protective equipment and the proper handling of tools and machinery.
2. **Hand Tools:** Students learn about the various hand tools used in the workshop, including hammers, screwdrivers, pliers, and saws. They also learn how to use these tools safely and effectively.
3. **Power Tools:** Students learn about the various power tools used in the workshop, including drills, grinders, and sanders. They also learn how to use these tools safely and effectively.
4. **Measuring and Marking:** Students learn about the various measuring and marking tools used in the workshop, including rulers, calipers, and squares. They also learn how to use these tools to make accurate measurements and marks.
5. **Materials:** Students learn about the various materials used in the workshop, including wood, metal, and plastic. They also learn about the properties of these materials and how to work with them.
6. **Joining Techniques:** Students learn about the various joining techniques used in the workshop, including welding, brazing, and soldering. They also learn how to use these techniques to join materials together.
7. **Finishing Techniques:** Students learn about the various finishing techniques used in the workshop, including sanding, painting, and polishing. They also learn how to use these techniques to give their projects a professional finish.

Overall, Workshop Practice Lab provides students with a solid foundation in the skills and techniques used in the workshop. By the end of the course, students should be able to safely and effectively use a range of tools and materials to complete basic workshop projects.

S. No. Mechanical Workshop Duration

1. **Introduction to Mechanical Workshop:** The duration of this workshop can vary depending on the depth of the topics covered. A basic introduction to mechanical workshop can take anywhere from a few hours to a full day.
2. **Basic Hand Tools and their Usage:** This workshop can take anywhere from half a day to a full day, depending on the number of tools covered and the level of detail in their usage.

3. Machine Tools and their Operation: The duration of this workshop can vary greatly depending on the number of machine tools covered and the level of detail in their operation. A basic introduction to machine tools can take anywhere from a full day to several days.
4. Welding and Fabrication: This workshop can take anywhere from a few days to a week, depending on the level of detail in the welding and fabrication techniques covered.
5. Computer-Aided Design and Manufacturing: The duration of this workshop can vary greatly depending on the software used and the level of detail in the design and manufacturing processes covered. A basic introduction to CAD/CAM can take anywhere from a few days to several weeks.
6. Robotics and Automation: This workshop can take anywhere from a few days to several weeks, depending on the level of detail in the robotics and automation technologies covered.
7. Quality Control and Inspection: The duration of this workshop can vary depending on the level of detail in the quality control and inspection techniques covered. A basic introduction to quality control and inspection can take anywhere from a few hours to a full day.
8. Maintenance and Repair: This workshop can take anywhere from a few days to a week, depending on the level of detail in the maintenance and repair techniques covered.
9. Health and Safety: The duration of this workshop can vary depending on the level of detail in the health and safety practices covered. A basic introduction to health and safety in a mechanical workshop can take anywhere from a few hours to a full day.
10. Project Management: This workshop can take anywhere from a few days to several weeks, depending on the level of detail in the project management techniques covered.

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

A mechanical workshop is a place where various materials, tools, and machines are used to manufacture, repair, or maintain mechanical systems. In this section, we will introduce some common materials, tools, and machines found in a mechanical workshop.

Materials

- **Metals:** Metals such as steel, aluminum, and copper are commonly used in mechanical workshops due to their strength, durability, and versatility.

- **Plastics:** Plastics are also commonly used in mechanical workshops due to their lightweight, low cost, and ease of fabrication.
- **Composites:** Composite materials, such as fiberglass and carbon fiber, are used in mechanical workshops for their high strength-to-weight ratio and corrosion resistance.

Tools

- **Hand tools:** Hand tools, such as hammers, screwdrivers, and wrenches, are used for manual tasks such as tightening bolts or driving nails.
- **Power tools:** Power tools, such as drills, saws, and grinders, are used for tasks that require more power or speed than can be achieved with hand tools.
- **Measuring tools:** Measuring tools, such as calipers, micrometers, and gauges, are used to measure dimensions and tolerances of mechanical parts.

Machines

- **Lathes:** Lathes are used to rotate a workpiece and remove material using cutting tools to create cylindrical shapes.
- **Milling machines:** Milling machines use rotary cutters to remove material from a workpiece to create a variety of shapes and features.
- **Drilling machines:** Drilling machines are used to create holes in a workpiece using a rotating drill bit.

These are just a few examples of the materials, tools, and machines that can be found in a mechanical workshop. A thorough understanding of these and other workshop equipment is essential for anyone working in the field of mechanical engineering.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop.

- **Layout:** The layout of a workshop refers to the arrangement of equipment, machinery, and workstations in a manner that maximizes efficiency, safety, and productivity. A well-designed layout takes into account the flow of materials and personnel, as well as the space and power requirements of the equipment.
- **Safety measures:** Safety measures in a workshop include the use of personal protective equipment (PPE) such as safety glasses, gloves, and hearing protection; proper training and supervision; and the implementation of safety procedures and protocols. Regular maintenance and inspection of equipment and machinery, as well as good housekeeping practices, can also help to prevent accidents and injuries.

- **Engineering materials:** Different engineering materials are used in a workshop depending on the specific application and requirements. Some common materials include:
 - **Mild steel:** A type of carbon steel with a low carbon content, mild steel is ductile, malleable, and easy to weld. It is commonly used for general-purpose applications such as construction and automobile manufacturing.
 - **Medium carbon steel:** With a higher carbon content than mild steel, medium carbon steel is stronger and harder but less ductile. It is often used for applications that require strength and toughness, such as shafts, gears, and machinery parts.
 - **High carbon steel:** High carbon steel has an even higher carbon content and is very hard and strong but brittle. It is commonly used for applications that require high strength and wear resistance, such as cutting tools and springs.
 - **High speed steel:** A type of tool steel, high speed steel is alloyed with elements such as tungsten, molybdenum, and cobalt to increase its hardness and heat resistance. It is commonly used for cutting tools such as drill bits and saw blades.
 - **Cast iron:** Cast iron is an alloy of iron, carbon, and silicon that is known for its excellent castability, wear resistance, and durability. It is commonly used for applications such as engine blocks, pipes, and machine frames.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section

Fitting, sheet metal, and welding are important processes in the manufacturing industry. These processes require the use of various tools, equipment, devices, and machines to achieve the desired results. Below is a list of some of the commonly used tools, equipment, devices, and machines in these sections:

1. **Fitting tools:** These include tools such as hammers, chisels, files, hack-saws, and hand reamers. These tools are used for cutting, shaping, and finishing metal parts.
2. **Sheet metal tools:** These include tools such as snips, punches, and shears. These tools are used for cutting and shaping sheet metal.
3. **Welding equipment:** This includes equipment such as welding machines, welding torches, and welding electrodes. These are used for joining metal parts together using heat and pressure.
4. **Measuring devices:** These include devices such as calipers, micrometers, and dial indicators. These devices are used for measuring the dimensions of metal parts.
5. **Machines:** These include machines such as lathes, milling machines, and drill presses. These machines are used for cutting, shaping, and drilling metal parts.

It is important to have a good understanding of these tools, equipment, devices, and machines in order to use them effectively and safely in fitting, sheet metal, and welding processes. Proper training and practice are essential for achieving proficiency in their use.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

1. **Vernier Calliper:** The least count of a Vernier calliper is the difference between the value of one main scale division and the value of one Vernier scale division. To calculate the least count, divide the value of one main scale division by the total number of Vernier scale divisions.
2. **Vernier Height Gauge:** The least count of a Vernier height gauge is calculated in the same way as that of a Vernier calliper. Divide the value of one main scale division by the total number of Vernier scale divisions.
3. **Micrometer (Screw Gauge):** The least count of a micrometer is the smallest distance that can be measured with the instrument. It is equal to the pitch of the screw, which is the distance between two adjacent threads on the screw.

To take different readings over given metallic pieces using these instruments, follow these steps: 1. Place the metallic piece between the jaws of the instrument. 2. Close the jaws gently until they touch the metallic piece. 3. Read the main scale and the Vernier scale or the thimble scale (in case of a micrometer) to obtain the measurement. 4. Repeat the process for different metallic pieces and record the readings.

Machine Shop (3 hours)

A machine shop is a facility where machining is performed. Machining is the process of removing material from a workpiece to create a desired shape or surface. This is typically done using machine tools such as lathes, milling machines, and drill presses. Here are some key points to consider when studying machine shops:

1. **Machine tools:** Machine shops typically contain a variety of machine tools, each designed for a specific type of machining operation. Common machine tools include lathes, milling machines, drill presses, and grinders.
2. **Workholding:** Workholding refers to the methods used to secure a workpiece in place while it is being machined. Common workholding methods include clamps, vises, and chucks.
3. **Cutting tools:** Cutting tools are used to remove material from the workpiece. They come in a variety of shapes and sizes, and are typically made from materials such as high-speed steel or carbide.

4. **Coolants and lubricants:** Coolants and lubricants are used to reduce heat and friction during the machining process. This can help to extend the life of the cutting tools and improve the surface finish of the workpiece.
5. **Safety:** Safety is an important consideration in any machine shop. Operators should be trained in the proper use of machine tools and should always wear appropriate personal protective equipment.
6. **Quality control:** Quality control is the process of ensuring that machined parts meet the desired specifications. This can involve the use of measuring tools such as micrometers and calipers, as well as more advanced techniques such as coordinate measuring machines.
7. **Maintenance:** Regular maintenance is important to keep machine tools in good working order. This can involve tasks such as cleaning, lubrication, and the replacement of worn or damaged parts.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

A lathe is a machine tool that rotates a workpiece about an axis of rotation to perform various operations such as cutting, sanding, knurling, drilling, deformation, facing, and turning. The lathe is one of the oldest and most important machine tools.

The main parts of a lathe machine are: - Headstock: It is the main part of the lathe machine that holds and rotates the workpiece. - Tailstock: It is used to support the other end of the workpiece. - Bed: It is the base of the lathe machine that supports all the other parts. - Carriage: It is used to move the cutting tool along the workpiece. - Cross slide: It is used to move the cutting tool perpendicular to the workpiece. - Compound rest: It is used to set the cutting tool at an angle to the workpiece. - Tool post: It is used to hold the cutting tool.

The main operations that can be performed on a lathe machine are: - Facing: It is the operation of machining the ends of a workpiece to produce a flat surface. - Plane Turning: It is the operation of machining a cylindrical surface on a workpiece. - Step Turning: It is the operation of machining different diameters on a workpiece. - Taper Turning: It is the operation of machining a conical surface on a workpiece. - Threading: It is the operation of cutting threads on a workpiece. - Knurling: It is the operation of producing a rough surface on a workpiece. - Parting: It is the operation of cutting a workpiece into two or more pieces.

These are the basic operations that can be performed on a lathe machine. There are many other operations and accessories that can be used to perform more complex tasks. It is important to understand the construction and working of a lathe machine to be able to use it effectively.

Fitting Shop (3 hours)

1. A fitting shop is a workshop where the process of fitting takes place.
2. Fitting is the process of assembling, shaping, and finishing parts to make them fit together accurately and precisely.
3. Fitting is an important step in the manufacturing process, as it ensures that the final product is of high quality and functions as intended.
4. In a fitting shop, workers use various tools and machines to shape and finish parts, such as lathes, milling machines, and grinders.
5. Fitting can involve processes such as cutting, drilling, tapping, reaming, and grinding.
6. Fitting is a skilled trade, and workers in a fitting shop must have a good understanding of materials, tools, and processes to produce high-quality work.
7. A fitting shop may be found in various industries, including manufacturing, construction, and engineering.
8. In a 3-hour session in a fitting shop, one could expect to learn about the tools and processes used in fitting, as well as gain hands-on experience in shaping and finishing parts.

1. Practice marking operations

- Marking operations are an essential part of many manufacturing processes.
- They are used to identify, label, and track parts and products.
- Marking operations can be performed using a variety of methods, including stamping, engraving, etching, and printing.
- The choice of marking method depends on factors such as the material being marked, the desired mark quality, and the production volume.
- It is important to practice marking operations to ensure that the marks are accurate, legible, and consistent.
- Practice can involve testing different marking methods and parameters to determine the best approach for a given application.
- Regular practice can also help to identify and address any issues with the marking equipment or process.
- In summary, practicing marking operations is important for achieving high-quality, consistent marks in manufacturing processes.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

The preparation of U or V -Shape Male Female Work piece involves several steps, including filing, sawing, drilling, and grinding. These steps are necessary to achieve the desired shape and fit of the work piece.

1. **Filing:** Filing is the process of removing excess material from the work piece using a file. This is done to smooth out rough edges and to shape the work piece to the desired form.

2. **Sawing:** Sawing is the process of cutting the work piece to the desired size and shape using a saw. This is typically done using a band saw or a hacksaw.
3. **Drilling:** Drilling is the process of creating holes in the work piece using a drill. This is necessary for the assembly of the work piece, as well as for the attachment of other components.
4. **Grinding:** Grinding is the process of removing excess material from the work piece using an abrasive wheel. This is done to achieve a smooth finish and to remove any burrs or rough edges.

These steps must be performed in the correct order and with the appropriate tools to achieve the desired result. Care must be taken to ensure that the work piece is properly aligned and secured during each step to prevent damage or inaccuracies.

4 Carpentry Shop 3 Hrs

Carpentry is a skilled trade and a craft in which the primary work performed is the cutting, shaping, and installation of building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc. Carpenters traditionally worked with natural wood and did the rougher work such as framing, but today many other materials are also used and sometimes the finer trades of cabinetmaking and furniture building are considered carpentry.

Here are some key points to consider when working in a carpentry shop for 3 hours:

1. **Safety:** Always wear appropriate safety gear, such as goggles, gloves, and ear protection. Follow all safety guidelines and procedures to prevent accidents.
2. **Tools:** Make sure you have all the necessary tools for the job, such as saws, hammers, and measuring tapes. Keep your tools organized and in good condition.
3. **Materials:** Choose the right materials for the job, such as wood, nails, and screws. Make sure you have enough materials to complete the job.
4. **Planning:** Plan your work carefully to make the most of your time in the shop. Make a list of tasks to complete and prioritize them.
5. **Technique:** Use proper techniques when cutting, shaping, and installing materials. Follow instructions and guidelines to ensure a high-quality result.
6. **Clean up:** Leave the shop clean and organized when you are finished. Put away tools and materials and sweep the floor.

By following these guidelines, you can make the most of your time in the carpentry shop and complete your work safely and efficiently.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

Carpentry involves the use of various tools and equipment to create wooden structures and objects. Some common carpentry tools and equipment include:

1. **Saws:** Used to cut wood to the desired size and shape. There are different types of saws, such as hand saws, circular saws, and jigsaws.
2. **Hammers:** Used to drive nails into wood to join pieces together.
3. **Chisels:** Used to shape and carve wood.
4. **Planes:** Used to smooth and shape wood.
5. **Squares:** Used to ensure that angles are accurate and that pieces are square.
6. **Clamps:** Used to hold pieces of wood together while they are being worked on.

There are several different types of joints that can be used in carpentry, including the cross half lap joint, the half lap dovetail joint, and the mortise and tenon joint.

- **Cross Half Lap Joint:** This joint is created by removing half of the thickness of each piece of wood where they will intersect, and then fitting the two pieces together. This creates a strong joint that is often used in framing and other structural applications.
- **Half Lap Dovetail Joint:** This joint is similar to the cross half lap joint, but it includes a dovetail shape that helps to lock the two pieces together. This joint is often used in furniture making and other applications where strength and appearance are important.
- **Mortise and Tenon Joint:** This joint is created by cutting a mortise (a hole) into one piece of wood and a tenon (a protruding piece) on the other piece of wood. The tenon is then inserted into the mortise, creating a strong joint. This joint is often used in furniture making and other applications where strength is important.

To make these joints, the carpenter must carefully measure and mark the wood, and then use the appropriate tools to cut and shape the pieces. Once the pieces are prepared, they can be assembled and glued or nailed together to create the final joint. It is important to work carefully and accurately to ensure that the joints are strong and well-made.

5 Welding Shop 6 Hrs

1. Welding is a fabrication process that joins two or more pieces of metal or thermoplastics by heating the surfaces to their melting points and applying

pressure to fuse them together.

2. Welding is used in a variety of industries, including construction, manufacturing, and repair.
3. There are several types of welding, including arc welding, gas welding, and resistance welding.
4. Welding safety is important and includes wearing protective gear, such as gloves, a welding helmet, and a respirator, and following proper procedures to prevent accidents.
5. A welding shop is a facility where welding services are provided. A 6-hour welding shop session may include instruction, practice, and the completion of a welding project.

Introduction to BI standards and reading of welding drawings

BI standards refer to the standards set by the British Institute of Standards for welding and fabrication. These standards provide guidelines for the design, fabrication, and inspection of welded joints.

Reading welding drawings is an essential skill for anyone involved in the welding and fabrication industry. Welding drawings provide detailed information about the dimensions, materials, and welding processes to be used in the fabrication of a component.

To practice reading welding drawings, it is important to understand the different types of joints and welding processes. Some common joints and welding processes include:

1. **Butt Joint:** A butt joint is a joint where two pieces of material are joined end to end. This type of joint is commonly used in welding and can be welded using various welding processes.
2. **Lap Joint:** A lap joint is a joint where two pieces of material overlap each other. This type of joint is commonly used in welding and can be welded using various welding processes.
3. **TIG Welding:** Tungsten Inert Gas (TIG) welding is a welding process that uses a non-consumable tungsten electrode to produce the weld. This process is commonly used for welding thin materials and produces high-quality welds.
4. **MIG Welding:** Metal Inert Gas (MIG) welding is a welding process that uses a consumable wire electrode to produce the weld. This process is commonly used for welding thicker materials and is faster than TIG welding.

By practicing these operations, you can improve your skills in reading welding drawings and understanding the different types of joints and welding processes.

6 Moulding and Casting Shop 6 Hrs

Moulding and casting are manufacturing processes used to produce complex shapes. These processes involve pouring molten metal into a mould, which contains a hollow cavity of the desired shape, and then allowing the metal to cool and solidify. The solidified part is then removed from the mould to complete the process.

1. Moulding is the process of creating a mould, which is a negative impression of the desired shape. This can be done using a variety of materials, including sand, plaster, and metal.
2. Casting is the process of pouring molten metal into the mould and allowing it to cool and solidify. The metal will take on the shape of the mould cavity, creating a replica of the desired shape.
3. Moulding and casting are used to produce a wide variety of parts, including engine blocks, gears, and decorative items.
4. The moulding and casting process can be done using a variety of metals, including iron, steel, aluminum, and copper.
5. There are several different types of casting processes, including sand casting, investment casting, and die casting.
6. The choice of casting process depends on several factors, including the size and complexity of the part, the desired surface finish, and the production volume.
7. Moulding and casting are versatile processes that can be used to produce parts with complex geometries and high levels of detail.
8. The moulding and casting process can be automated, allowing for high levels of production efficiency and consistency.
9. Moulding and casting are widely used in a variety of industries, including automotive, aerospace, and consumer goods.
10. The moulding and casting process requires specialized knowledge and equipment, and is typically done by skilled professionals in a dedicated casting shop.

Introduction to Patterns, Pattern Allowances, Ingredients of Moulding Sand and Melting Furnaces. Foundry Tools and Their Purposes Demo of Mould Preparation and Aluminum Casting Practice – Study and Preparation of Mould for Plastic

1. **Patterns:** A pattern is a model or replica of the object to be cast. It is used to create the cavity in the mould where the molten metal will be poured to form the casting.

2. **Pattern Allowances:** Pattern allowances are small adjustments made to the pattern to compensate for changes that occur during the casting process. These include shrinkage allowance, draft allowance, machining allowance, and distortion allowance.
3. **Ingredients of Moulding Sand:** Moulding sand is a mixture of sand, clay, and water. The sand provides the structure for the mould, the clay binds the sand together, and the water helps to make the sand more workable.
4. **Melting Furnaces:** Melting furnaces are used to melt the metal that will be used in the casting process. There are several types of melting furnaces, including cupola furnaces, induction furnaces, and electric arc furnaces.
5. **Foundry Tools:** Foundry tools are used to prepare the mould and handle the molten metal. These include trowels, rammers, sprue cutters, and ladles.
6. **Demo of Mould Preparation and Aluminum Casting:** In this demo, a mould is prepared using a pattern and moulding sand. The molten aluminum is then poured into the mould and allowed to cool and solidify. The casting is then removed from the mould and any excess material is trimmed away.
7. **Practice – Study and Preparation of Mould for Plastic:** In this practice, students will study the process of preparing a mould for plastic casting. They will then prepare their own mould and use it to cast a plastic object.

7 CNC Shop 6 Hrs

1. CNC stands for Computer Numerical Control.
2. A CNC shop is a workshop where CNC machines are used to manufacture parts and products.
3. CNC machines are controlled by computer programs, which allow for precise and accurate cutting, drilling, and shaping of materials.
4. A 6-hour shift in a CNC shop would involve operating and monitoring the CNC machines, loading and unloading materials, and performing quality checks on finished products.
5. CNC shops can produce a wide variety of products, from small, intricate parts to large, complex structures.
6. CNC technology has revolutionized the manufacturing industry, allowing for faster production times and increased efficiency.
7. Working in a CNC shop requires technical knowledge and skills, as well as attention to detail and the ability to follow instructions and procedures.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

CNC (Computer Numerical Control) machines are automated machines that are

capable of performing a variety of machining operations on metal components. These machines are controlled by computer programs and can be used to perform operations such as drilling, milling, turning, and grinding.

Some of the main features and working parts of a CNC machine include:

1. **Control Panel:** The control panel is the interface between the operator and the machine. It is used to input commands and control the operation of the machine.
2. **Spindle:** The spindle is the part of the machine that holds and rotates the cutting tool. It is driven by a motor and can rotate at high speeds.
3. **Cutting Tool:** The cutting tool is the part of the machine that removes material from the workpiece. There are many different types of cutting tools that can be used, depending on the operation being performed.
4. **Worktable:** The worktable is the part of the machine that holds the workpiece. It can move in multiple directions to position the workpiece for machining.
5. **Coolant System:** The coolant system is used to cool the cutting tool and workpiece during machining. This helps to prevent overheating and prolong the life of the cutting tool.

There are many different accessories that can be used with a CNC machine to perform different operations on metal components. Some common accessories include:

1. **Vises:** Vises are used to hold the workpiece securely in place during machining.
2. **Collets:** Collets are used to hold the cutting tool securely in the spindle.
3. **End Mills:** End mills are a type of cutting tool that can be used for milling operations.
4. **Drill Bits:** Drill bits are a type of cutting tool that can be used for drilling operations.

By using the appropriate accessories and cutting tools, a CNC machine can be used to perform a wide variety of operations on metal components. These operations can include drilling, milling, turning, and grinding, among others.

8 To prepare a product using 3D printing 3 Hrs

1. **Design the 3D model:** The first step in preparing a product using 3D printing is to design a 3D model of the product using computer-aided design (CAD) software. This model will serve as the blueprint for the 3D printer to create the physical product.

2. **Choose the right 3D printing technology:** There are several 3D printing technologies available, each with its own advantages and limitations. The most common technologies are Fused Deposition Modeling (FDM), Stereolithography (SLA), and Selective Laser Sintering (SLS). The choice of technology will depend on the material, size, and complexity of the product.
3. **Prepare the 3D printer:** Before starting the printing process, the 3D printer must be prepared by loading the appropriate material, calibrating the print bed, and setting the printing parameters.
4. **Print the product:** Once the 3D printer is prepared, the printing process can begin. The 3D printer will read the 3D model and build the product layer by layer.
5. **Post-processing:** After the printing process is complete, the product may require post-processing to remove support structures, smooth the surface, or apply a finish.
6. **Quality control:** The final step in preparing a product using 3D printing is to perform quality control to ensure that the product meets the desired specifications and standards.

In summary, preparing a product using 3D printing involves designing a 3D model, choosing the right 3D printing technology, preparing the 3D printer, printing the product, post-processing, and performing quality control. This process can take approximately 3 hours, depending on the size and complexity of the product.

Total 33 Hrs

1. The topic “Total 33 Hrs” is not clear. Could you please provide more information or context about the topic?
2. Is “Total 33 Hrs” related to a specific subject or event?

Course Outcome:

- A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to have acquired by the end of a course.
- Course outcomes are used to assess the effectiveness of a course in achieving its intended goals.
- Course outcomes are typically written in measurable terms, using action verbs that describe what students should be able to do by the end of the course.
- Course outcomes should be aligned with the overall goals and objectives of the program or curriculum in which the course is offered.
- Course outcomes can be used to guide the development of course content, assessments, and teaching strategies.

- Course outcomes can also be used to communicate expectations to students and to provide a basis for evaluating student learning.
- Course outcomes should be reviewed and revised periodically to ensure that they remain relevant and up-to-date.

The students will be able to:

- CO1: Use various engineering materials, tools, machines and measuring equipment at a K3 Bloom's level.
- CO2: Perform machine operations in lathe and CNC machine at a K3 Bloom's level.
- CO3: Perform manufacturing operations on components in fitting and carpentry shop at a K3 Bloom's level.
- CO4: Perform operations in welding, moulding, casting and gas cutting at a K3 Bloom's level.
- CO5: Fabricate a job by 3D printing manufacturing technique at a K3 Bloom's level.

Reference Books:

- Reference books are a type of non-fiction book that provides information on a specific subject.
- They are often used for research or to gain a deeper understanding of a topic.
- Examples of reference books include encyclopedias, dictionaries, thesauruses, almanacs, and atlases.
- Reference books are typically organized in a way that makes it easy to find the information you are looking for.
- They are often considered authoritative sources of information and are frequently updated to ensure accuracy.
- Reference books can be found in libraries, bookstores, and online.
- It is important to evaluate the credibility of a reference book before using it as a source of information.

Workshop Practice by H S Bawa, published by McGraw Hill

- Workshop Practice is a book written by H S Bawa and published by McGraw Hill.
- The book covers various topics related to workshop technology and practices.
- It is a comprehensive guide for students and professionals in the field of engineering and technology.
- The book provides detailed information on various workshop tools, machines, and processes.
- It also includes practical examples and illustrations to help readers understand the concepts better.

- The book is well-organized and written in a clear and concise manner, making it easy to follow and understand.
- Overall, Workshop Practice by H S Bawa is a valuable resource for anyone interested in learning more about workshop technology and practices.

2. Mechanical Workshop Practice, K C John, PHI

- Mechanical Workshop Practice is a book written by K.C. John and published by PHI Learning in 2010 .
- The book has an ISBN 10 of 812034166X and an ISBN 13 of 9788120341661 .
- The book describes the general principles of different workshop processes such as primary and secondary shaping processes, metal joining methods, surface finishing, and heat treatment .
- The workshop processes are explained with the help of diagrams and photographs .
- The book is intended for undergraduate students of mechanical engineering and related disciplines .

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a two-volume book set written by HazraChoudhary and published by Media Promoters and Publications.
- The books provide a comprehensive guide to workshop practices and techniques.
- Volume 1 covers topics such as workshop safety, hand tools, measuring instruments, and machine tools.
- Volume 2 delves into more advanced topics such as welding, sheet metal work, and heat treatment.
- The books are written in a clear and concise manner, making them suitable for both beginners and experienced practitioners.
- The books are widely used as reference material in technical schools and colleges.
- The books are well-illustrated with diagrams and photographs to aid understanding.
- The books are regularly updated to keep pace with the latest developments in workshop practice.

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

CNC stands for Computer Numerical Control. It is a technology that uses computer-controlled machines to perform various manufacturing operations. CNC machines are widely used in industries such as aerospace, automotive, and electronics.

Some of the key concepts covered in this book include:

1. Introduction to CNC machines: This section provides an overview of the different types of CNC machines, their components, and their applications.
2. CNC programming: This section covers the basics of CNC programming, including G-code and M-code, and provides examples of CNC programs.
3. CNC tooling: This section discusses the different types of tooling used in CNC machines, including cutting tools, tool holders, and workholding devices.
4. CNC operations: This section covers the various operations that can be performed on a CNC machine, including milling, turning, drilling, and tapping.
5. CNC machine maintenance: This section discusses the importance of regular maintenance of CNC machines and provides guidelines for maintaining and troubleshooting CNC machines.

Overall, this book provides a comprehensive introduction to the fundamentals of CNC technology and its applications in manufacturing. It is a valuable resource for students, engineers, and technicians who are interested in learning more about CNC machines and their programming.